# Infant Mortality Rates in the U.S.

Super Stars

## Introduction

Infant mortality rate is an important indicator of the health of a society (Centers for Disease Control and Prevention, 2023). The U.S. is lagging behind in international rankings for infant mortality rates, and there needs to be more change implemented to keep pace with other respective developed countries (Singh & Yu, 2019). Therefore, our population of interest are infants in the U.S.. We pose the following three research questions: **(1) Do Democratic states have a lower infant mortality rate than Republican states? (2) Do states in the Western regions of the U.S. have a lower infant mortality rate than states in the Northeast (3) Do states with a healthcare system performance ranked in the top 25 of all U.S. states have a significantly lower infant mortality rate than states ranked with a healthcare system performance in the bottom 25?** Our research questions are relevant and novel because they look at variables that specifically relate to the structure and function of each U.S. state, which will provide insights into the broader factors that influence infant mortality rate. Most of our data was collected through the National Vital Statistics Reports of 2021 (NVSS), Pew Research Center, and the 2022 Scorecard on State Health System Performance produced by The Commonwealth Fund. We had to exclude Vermont from our analysis because there was insufficient data on Vermont's infant mortality rate in the NVSS reports. We had 50 observations in total - 49 U.S. States plus D.C..

## Methods and Analysis

**EDA & Assessing Multicollinearity.** We first confirmed that our response variable, infant mortality rate, is appropriate for regression because it is continuous, unimodal, approximately symmetric, and does not contain extreme outliers. The quantitative variables unmarried_all (r=0.590), low_birthw_all (r=0.697), and rate_twins (r=0.350) (Figures 2,3,4) had the highest correlation coefficients. They all had a weak to moderately strong positive linear relationship with infant mortality rate. The qualitative variable health_system_performance had the strongest relationship with our response (Figure 1). "Top 25" states had a lower mean infant mortality rate compared to "Bottom 25" states. Lastly, there appeared to be two significant interactions between region and political_affiliation and health_system_performance and political_affiliation. There was no evidence of severe multicollinearity. All individual VIFs were below 10, and the average VIF was 1.844, which is less than 3. There were also no strong patterns in the scatterplots of the pairwise relationships (Figure 5).

**Model Building Through Multiple Linear Regression (MLR).** We first performed stepwise regression with a pent of 0.1 and a prem of 0.3. Stepwise regression selected low_birtw_all, rate_twins, and unmarried_all. We then added our qualitative variables to the model. Using a significance level of 0.05, we performed a t-test for health_system_performance, and a nested F-test for political_affiliation and region. The variable health_system_performance was the only significant qualitative variable. However, unmarried_all became insignificant when health_system_performance was added to the model and thus, unmarried_all was removed. Lastly, we checked the two interactions mentioned in our EDA. However, neither was significant in the nested F-tests. We arrived at our final model that included the variables rate_twins, low_birthw_all, and health_system_performance.

**Assumptions & Influential Diagnostics.** Our final model met all four assumptions because there were no obvious patterns in the residual plots, no obvious fanning patterns in the residual vs. fitted plot, most residuals fell along the QQ-line, and we did not work with time series data. We identified four influential observations and outliers: observations 2, 24, 31, and 34. In an effort to retain the greatest amount of data yet produce the "best" model, we removed observation 2 (Alaska) and re-fitted our model. This improved our Adjusted R^2 value from 0.57 to 0.65. There were no other variable transformations that significantly improved our model or corrected for the outlier points.

**Model Building Through Ridge Regression (RR).** RR is an alternative to MLR. Although RR is typically used when there is severe multicollinearity in the data, the paper by Brook and Moore show that least squares regression coefficients are generally much further away from the true coefficients and that this fact is independent of multicollinearity (Brook & Moore, 1980). Thus, even if there is no concern for multicollinearity it is advantageous to shrink the least squares regression coefficients to get more accurate ones (Brook & Moore, 1980). Because "glmnet" does not calculate t-values when doing RR, we got another R code to do the RR variable selection. Our final RR model selected the following variables: rate_twins, unmarried_all, low_birthw_all, Democratic state, Northeast region, and Southeast region.

## Results

Our final MLR model had a global F-statistic of 30.08 on 3, 45 degrees of freedom and a p-value $< 0.0001$, indicating that our model is adequate. Our model had an Adjusted R^2 value of 0.6451, implying that about 64.51% of the variation in infant mortality rate

is explained by our model. The model also had a relatively low RMSE value of 0.79, which means that approximately 95% of our predictions will be within 2*0.79=1.58% of the actual mortality rate for U.S. states. Lastly, the confidence intervals for all our beta estimates were relatively narrow and none included zero, confirming that all the explanatory variables in our model are significant. Thus, overall, our model is moderately good. Our final model was: $E(InfantMortalityRate) = \beta_0 + \beta_1 RateTwins + \beta_2 LowBw + \beta_3 HealthSystemPerformanceTop25$where {health_system_performanceTop25 = 1 if "Top 25"; 0 if "Bottom 25"}

## Conclusions

**MLR Model.** Our research questions focused on political_affiliation, region, and health_system_performance. MLR showed that political_affiliation and region were not significant predictors for infant mortality rates, but health system performance was, along with rate_twins and low_birthw_all. Our final prediction equation was: E(Infant Mortality Rate)=-3.8334+0.1327*rate_twins+0.6446*low_birthw_all-0.6040*health_system_performanceTop25 (where health_system_performanceTop25 = 1 if "Top 25"; 0 if "Bottom 25"). For a 1% increase in the birth rate for twins, the expected infant mortality rate increases by 0.1327, given that the other variables are held constant. For a 1% increase in the percentage of low birth weight infants, the expected infant mortality rate increases by 0.6446, given that the other variables are held constant. Lastly, the mean infant mortality rate for states with a health system performance in the top 25 is 0.6040% lower than the infant mortality rate for states with a health system performance in the bottom 25. Our MLR model was used to predict the 2020 infant mortality rate in West Virginia. The model predicted 9.047, whereas the actual value is 7.33. The prediction did not fall within the 2*RMSE threshold, but is also not too far off the threshold. This reflects that our model is moderately adequate, but still needs to be improved.

**RR Model.** One of the first steps we took to produce a more accurate model was through RR. Our final RR prediction equation was: E(Infant Mortality Rate)= -1.6414+0.0843*RateTwins+0.0471*Unmarried+0.3432*LowBw-0.5395*Democratic-0.6660*Northeast+0.3763*Southeast. Unlike our MLR model, the RR model suggests that a 1% increase in unmarried_all increases the expected infant mortality rate, democratic states and Northeastern states have a lower mean infant mortality rate than other states, while Southeastern states have a higher mean infant mortality rate than other states. Our RR model did a better job of predicting the 2020 infant mortality rate in West Virginia. RR predicted a mortality rate of 8.497, which is closer to the actual value of 7.33.

**Limitations and Future Improvements.** Some areas of improvement include adding more variables such as maternal age and socioeconomic status to our model to find better predictors of infant mortality rate. We can also collect infant mortality data from other years and build their respective prediction models to confirm the significance of selected variables. A limitation of the data was that the infant mortality rate for the state of Vermont was missing; therefore, the model was not representative of the entire United States. We also had to remove the state of Alaska from our dataset since it was an influential observation. Thus, the infant mortality rate in Alaska will need to be further studied individually to understand why it is an outlier.

## Appendix A: Data Dictionary

| Variable Name | Abbreviated Name | Description |
|---|---|---|
| State name | state | Name of a U.S. state, for which the infant mortality rate was collected. We collected data on 49 U.S. states (excluding Vermont), and D.C.. Each observation in our data table represents a U.S. state. We have 50 observations in total. |
| Infant Mortality Rate (%) (response) | mortality_rate | This variable is referred to as infant mortality rate, or just mortality rate, in our report. It represents the 2021 infant mortality rate for each U.S. state and D.C.. Infant mortality rate is defined as the number of infant deaths for every 1,000 live births. |
| Region of the U.S. | region | This variable is referred to as region in our report. We divide the 49 U.S. states and D.C. into five distinct geographical regions in the U.S., creating five levels for this variable: Midwest, Northeast, Southwest, West, and Southwest. |
| Rate of Pre-Pregnancy Diabetes | pp_diabetes | This variable refers to the number of births to mothers with pre-pregnancy diabetes per 1,000 births in 2021, for each U.S. state and D.C.. |
| Percentage of At-Home Births (%) | births_at_home | This variable refers to the percentage of births that occurred at home in 2021, for each U.S. state and D.C.. |
| Birth Rate for Twins | rate_twins | This variable refers to the birth rate for twins per 1,000 births for each U.S. state and D.C. in 2021. |

| Variable Name | Abbreviated Name | Description |
|---|---|---|
| Birth Rate for Triplets or Higher-Order Births | rate_triplets_higherorder | This variable refers to the birth rate for triplets or higher-order births per 1,000 births for each U.S. state and D.C. in 2021. |
| Percentage of Births to Unmarried Mothers (%) | unmarried_all | This variable refers to the percentage of births occurring to unmarried mothers in each U.S. state and D.C. in 2021. |
| Percentage of Low Birth Weight Infants (%) | low_birthw_all | This variable refers to the percentage of births that are low birth weight in each U.S. state and D.C. in 2021. Low birth weight is defined as less than 1,500 grams. |
| Political Affiliation | political_affiliation | This variable is referred to as political affiliation in our report. It has three levels: Democratic, Republican, and Split (where "Split" indicates that the state is "split" between both parties and does not have a democratic or republican lean). It represents the 2024 majority political affiliation of each U.S. State. This is determined based on both the legislative majority and party affiliation of state leaders, as well as the affiliation of the majority of the public. |
| Health System Performance | health_system_performance | This variable is referred to as health system performance in our report. States are classified into two levels: Top 25 and Bottom 25, where Top 25 indicates higher than average health performance and Bottom 25 indicates lower than average health performance. Rankings are based on the 2022 State Scorecard Rankings of overall performance. |

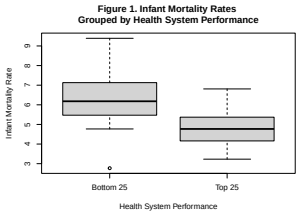
## Appendix B: Data Rows

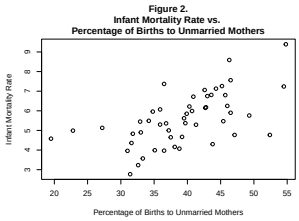
| | state | mortality_rate | region | pp_diabetes | births_at_home | rate_twins |
|---|---|---|---|---|---|---|
| 1 | Alabama | 7.56 | Southeast | 13.3 | 0.7 | 35.4 |
| 2 | Alaska | 7.37 | West | 13.6 | 2.5 | 30.3 |
| 3 | Arizona | 5.47 | Southwest | 12.6 | 1.4 | 29.0 |
| 4 | Arkansas | 8.59 | Southeast | 10.6 | 1.2 | 31.8 |
| 5 | California | 4.07 | West | 9.0 | 1.0 | 29.2 |
| 6 | Colorado | 4.99 | West | 8.9 | 1.9 | 30.4 |

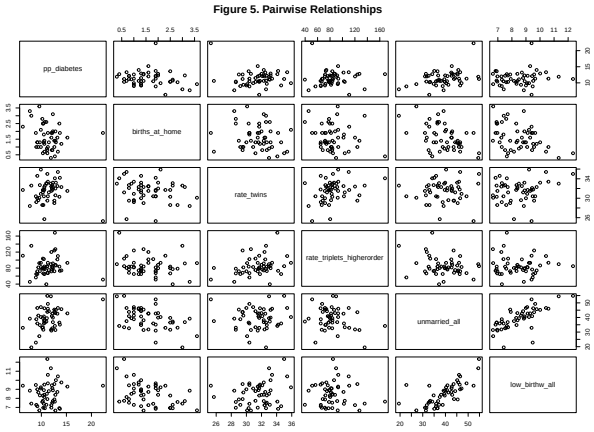
| | rate_triplets_higherorder | unmarried_all | low_birthw_all | political_affiliation |
|---|---|---|---|---|
| 1 | 110.1 | 46.5 | 10.43 | Republican |
| 2 | 83.7 | 36.5 | 6.92 | Republican |
| 3 | 72.1 | 45.4 | 7.88 | Republican |
| 4 | 64.9 | 46.3 | 9.52 | Republican |
| 5 | 65.2 | 38.8 | 7.28 | Democratic |
| 6 | 56.6 | 22.8 | 9.48 | Democratic |

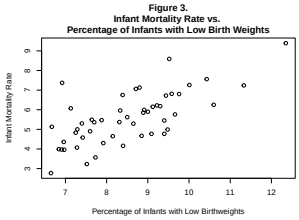
| | health_system_performance |
|---|---|
| 1 | Bottom 25 |
| 2 | Bottom 25 |
| 3 | Bottom 25 |
| 4 | Bottom 25 |
| 5 | Top 25 |
| 6 | Top 25 |

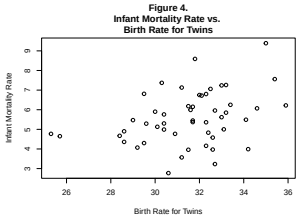
# Appendix C: Tables and Figures

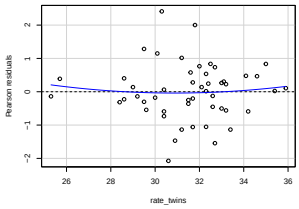


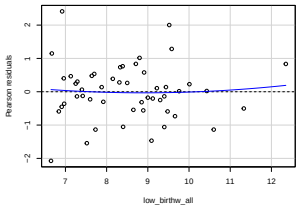



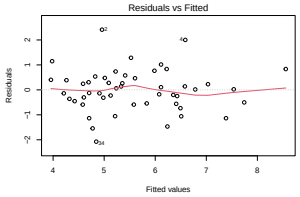



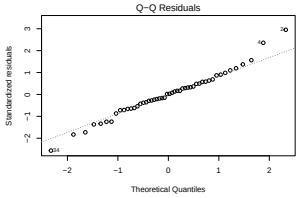



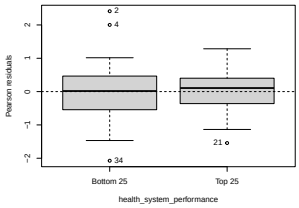



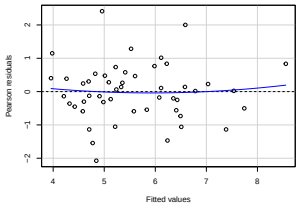



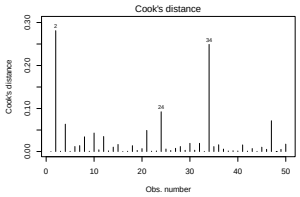



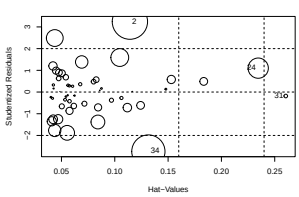



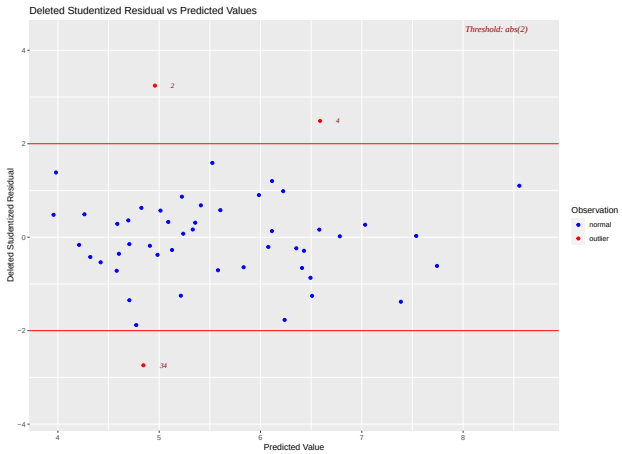




| | StudRes | Hat | CookD |
|---|---|---|---|
| 2 | 3.2438471 | 0.1141934 | 0.280964312 |
| 24 | 1.1008740 | 0.2345206 | 0.092398835 |
| 31 | -0.1832205 | 0.2603626 | 0.003017653 |
| 34 | -2.7406139 | 0.1315094 | 0.249077979 |




## Final MLR Model Ouput

```
Call:
lm(formula = mortality_rate ~ rate_twins + low_birthw_all + health_system_performance,
    data = clean_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.74285 -0.37939 -0.02637  0.47861  2.06789

Coefficients:
                                Estimate Std. Error t value Pr(>|t|)
(Intercept)                     -3.83393    1.82760  -2.098   0.0416 *
rate_twins                       0.13268    0.05356   2.477   0.0171 *
low_birthw_all                   0.64461    0.10480   6.151 1.86e-07 ***
health_system_performanceTop 25 -0.60402    0.25611  -2.358   0.0228 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.79 on 45 degrees of freedom
Multiple R-squared:  0.6673,    Adjusted R-squared:  0.6451
F-statistic: 30.08 on 3 and 45 DF,  p-value: 7.927e-11
```

## RR Model Output

***Supplemental information:**

RR is done by introducing a constant to modify the MLR solution. This constant is also called a "tuning parameter" and denoted by k (called "lambda" in the R "glmnet" package). There are two well-known values of k that can be selected from the MLR regression. One was found by Hoerl, Kennard and Baldwin (Hoerl et al., 1976), denoted k_HKB. The other was found by Lawless and Wang (Lawless & Wang, 1976), denoted k_LW.

0.06441878 is the Lawless and Wang (1976) tuning parameter we used for RR. It is the reciprocal of the F-statistic in the least squares regression.

RR variable selection is done as follows: a) First a full MLR regression is fit to the data; b) Using either k_HKB or k_LW the t-values for each coefficient is calculated; c) Using a significance level of 15%, only significant variables are collected; and d) the significant variables collected are used to refit RR (using a new k_HKB or k_LW), which is the final model.

```
[1] 0.06441878
```

| | RR beta | t-value | sig=1 | VIF | RMSE | LS beta |
|---|---|---|---|---|---|---|
| intercept.RR | -6.04165979 | 9999.000000 | 9999 | 9999.0000000 | 0.9065808 | -6.7583306 |
| X2 | 0.23061436 | 1.342343 | 1 | 1.1007212 | 9999.0000000 | 0.2802183 |
| X3 | 0.15600347 | 2.690161 | 1 | 0.9409146 | 9999.0000000 | 0.1647766 |
| X5 | 0.05594152 | 2.526120 | 1 | 1.5415614 | 9999.0000000 | 0.0578531 |
| X6 | 0.48590400 | 3.819348 | 1 | 1.5280024 | 9999.0000000 | 0.5193898 |

| | LS t | LS VIF | LS RMSE |
|---|---|---|---|
| intercept.RR | 9999.000000 | 9999.000000 | 0.9048456 |
| X2 | 1.485207 | 1.327550 | 9999.0000000 |
| X3 | 2.618952 | 1.107578 | 9999.0000000 |
| X5 | 2.158436 | 2.258268 | 9999.0000000 |
| X6 | 3.388020 | 2.218688 | 9999.0000000 |

## Appendix D: References

### Background


1. Centers for Disease Control and Prevention. (2023, September 13). *Infant Mortality.* Reproductive Health. https://www.cdc.gov/reproductivehealth/maternalinfanthealth/infantmortality.htm#
2. Okobi, O.E., et al. (2023). Trends and Factors Associated With Mortality Rates of Leading Causes of Infant Death: A CDC Wide-Ranging Online Data for Epidemiologic Research (CDC WONDER) Database Analysis. *Cureus*, *15*(9). https://doi.org/10.7759/cureus.45652
3. Singh, G. K., & Yu, S. M. (2019). Infant mortality in the United States, 1915-2017: Large social inequalities have persisted for over a century. *International journal of MCH and AIDS*, *8*(1), 19-31. https://doi.org/10.21106/ijma.271


### Data


1. Ely, D., & Driscoll, A. (2023). Infant Mortality in the United States, 2021: Data From the Period Linked Birth/Infant Death File. *National Vital Statistics Reports*, *73*(11). https://dx.doi.org/10.15620/cdc:131356

2. Gregory, E., & Ely, D. (2023). Trends and characteristics in prepregnancy diabetes: United States, 2016–2021. *National Vital Statistics Reports*, *73*(6). https://doi.org/10.15620/cdc:127679

3. Osterman, M., et al. (2023). Births: Final Data for 2021. *National Vital Statistics Reports*, *73*(1). https://dx.doi.org/10.15620/cdc:122047

4. Pew Research Center. (2024). *Religious landscape study*. Pew Research Center's Religion & Public Life Project. https://www.pewresearch.org/religion/religious-landscape-study/compare/party-affiliation/by/state/

5. Radley, D. C., Baumgartner, J. C., & Collins, S. R. (2022, June 16). 2022 Scorecard on State Health System Performance. https://www.commonwealthfund.org/publications/scorecard/2022/jun/2022-scorecard-state-health-system-performance

**Supplemental Code and Analysis Help**


1. Brook, R.J., & Moore, T. (1980). On the expected length of the least squares coefficient vector (Vol 12., pp. 245-246), Journal of Econometrics.
2. Hoerl, A. E., & Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems (Vol 12., pp. 55-67). Technometrics.
3. Hoerl, A. E., Kennard, R. W., & Baldwin, K. F. (1976). Ridge regression: Some simulations (Vol. A5, pp. 77-88 ). Communications in Statistics.
4. Hoerl, R. W., Schuenemeyer, J. H., & Hoerl, A. E. (1986). A simulation of biased estimation and subset selection regression techniques (Vol 28(4), pp. 369-380). Technometrics.
5. Lawless, J. F., & Wang, P. (1976). A simulation study of ridge and other regression estimators (Vol 5(4), pp. 307-323). Communications in Statistics - Theory and Methods.
6. Mendenhall, W., & Sincich, T. (2012). A Second Course in Statistics – Regression Analysis. Prentice Hall.